Grading Rubric

This is the scoring logic for every product audit. Apply it consistently.

The five tiers

Tier	Meaning	Trigger
Puberty Safe	All ingredients clear, no flags	Zero flagged ingredients in the reference PDF, OR only `LOW TOX` ingredients + naturally occurring fragrance components
Use Caution	Minor flags only, generally usable but not ideal	1–2 single-flag ingredients, none in heavy metal / formaldehyde-releaser / paraben / fragrance categories
Avoid (Mild)	Multiple flags or one significant flag	3+ flagged ingredients, OR any single ingredient with 3+ risk categories, OR any paraben
Avoid (Moderate)	Significant concerns across multiple categories	Heavy metals (cat. 3) present, OR formaldehyde-releaser, OR fragrance/parfum without disclosure, OR 5+ flagged ingredients
Avoid (High Priority Swap)	Multiple high-concern flags stacking	Heavy metals + endocrine disruptors + fragrance, OR 8+ flagged ingredients, OR known carcinogen (cat. 1) in a leave-on product, OR PFAS-treated ingredient in any product

Risk category reference (from PDF)

1. Carcinogen
2. Endocrine Disruption
3. Heavy Metals (bio-accumulation)
4. Allergies, Irritation, Corrosion
5. Microbiome-Busting (skin barrier)
6. Microplastics (non-biodegradable)
7. Acutely Toxic
8. Respiratory Irritant, asthmagen

Scoring procedure (run for every product)

1. **Pull the verbatim ingredient list** from the brand's product page into the spreadsheet's

`ingredients_verbatim` column.

2. **For each ingredient, look it up in `Ingredient_Safety_Reference.pdf`:**
   - If `LOW TOX`: no flag
   - If listed with risk numbers: record the numbers
   - If not in the PDF but matches a watched naming pattern (`-nium`, `-eth`, `-paraben`, `-glycol`, `Fragrance/Parfum`, `-siloxane`, `-sulfate`, formaldehyde-releaser names): flag it using the pattern's typical risk categories and note `inferred from naming pattern`
   - If neither in the PDF nor matching a pattern: log as `unknown — manual review needed` in the unknown_ingredients column
3. **Separate naturally occurring fragrance components.** These are flagged but soft — record them in a separate column (`naturally_occurring_allergens`) rather than the main flagged_ingredients column:
   - Linalool, limonene, citral, citronellal, geraniol, eugenol, isoeugenol, coumarin, farnesol, hydroxycitronellal, benzyl alcohol (when from essential oil), bisabolol
   - Only treat as soft if the product also lists the source essential oil (e.g., lavender oil, citrus oil, rose oil). If they appear standalone with no carrier oil, treat as full fragrance flags.
4. **Apply conditional rules** (see below).
5. **Tally the flags** and assign a tier per the tier table.
6. **Write a one-line summary** in the brand's voice for the report (NOT the consumer voice from the Scanner — neutral analytical voice for internal use).

Conditional rules — these override surface flags

These come straight from the Ingredient Safety Reference. Always check before finalizing.

Product-type conditionals

Ingredient	Rule
Titanium dioxide (CI 77891)	Not a concern in cream-based skincare or makeup; flag in sprays, aerosols, powders, oral care
Mica	Not a concern unless it's the first or main ingredient
Zeolite	Avoid in powdered form only

Ingredient	Rule
Zinc oxide (non-nano)	Low tox in creams/lotions; avoid in sprays and aerosols
Acetic acid (vinegar)	Flag only if used in a way that creates inhalation exposure
Enzymes (amylase, cellulase, subtilisin)	Flag in powdered products; lower concern in liquids

Hair-product specific

This is critical because Innersense and similar brands are hair-focused.

- **Conditioners and leave-ins**: Quaternary ammonium compounds (`-nium` pattern) are extremely common as conditioning agents. Flag them, but in the report note that quat-free conditioners are rare. This is context for the content angle, not a reason to soften the grade.
- **Behentrimonium methosulfate**: Marked LOW TOX in the reference despite the `-nium` pattern. Do not flag.
- **Cetearyl alcohol, cetyl alcohol, stearyl alcohol, behenyl alcohol, brassica alcohol**: These are fatty alcohols, not drying alcohols. LOW TOX. Do not flag.
- **Curl creams, gels, stylers**: These are leave-on products. Apply leave-on tier thresholds (more conservative).
- **Shampoos**: Rinse-off but daily. Sulfate-flagged ingredients still trigger an Avoid grade because of cumulative exposure.

Fragrance/Parfum specific

- "Fragrance" or "Parfum" listed without a breakdown = full flag (cat. 1, 2, 4, 8).
- "Fragrance (proprietary blend of essential oils)" with the EOs disclosed elsewhere on the page = soft flag.
- "Natural Fragrance" without disclosure = full flag. Do not treat "natural" as protective.
- Individual EOs listed (lavender oil, rose oil, etc.) without "fragrance" wrapper = no fragrance flag, but check for naturally occurring allergens.

Heavy metals — automatic escalation

If any ingredient in category 3 (heavy metals) appears, the minimum tier is **Avoid (Moderate)**. If a heavy metal stacks with any cat. 1 or cat. 2 flag, escalate to **Avoid (High Priority Swap)**.

PFAS-treated ingredients — automatic escalation

Any ingredient noted as "PFAS treated" in the reference triggers a minimum tier of **Avoid (High**

Priority Swap) regardless of other flags.

Formaldehyde-releasers — automatic escalation

DMDM hydantoin, diazolidinyl urea, imidazolidinyl urea, quaternium-15, methenamine, paraformaldehyde, benzylhemiformal, 5-bromo-5-nitro-1,3-dioxane, sodium hydroxymethylglycinate, glyoxal, polyoxymethylene urea, 2-bromo-2-nitropropane-1,3-diol → minimum tier **Avoid (Moderate)**.

Output requirements

Every product gets:

- A tier (one of the five)
- A flag count
- A list of flagged ingredients with their risk categories
- A list of naturally occurring allergens (separate)
- A list of unknown ingredients
- A one-sentence neutral summary

See `OUTPUT_TEMPLATES.md` for exact column structure.